

Initiating Search

November 4, 2025, 11:21 AM

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CAS Lexicon Search:

Kumada coupling reaction - Preferred Concept

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Exported: Retrieved Related Reaction Results (2,586)	 Reactions	View Results

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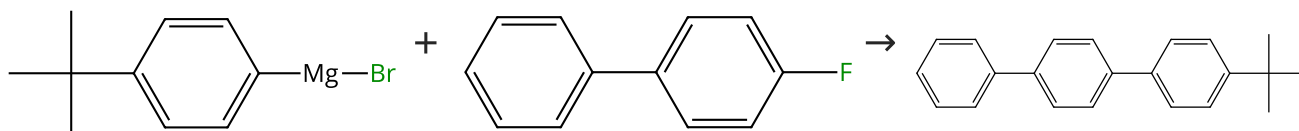
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (30)

 Suppliers (76)

 Suppliers (11)

31-176-CAS-22542818

Steps: 1 Yield: 100%

Aryl Fluoride Activation through Palladium-Magnesium Bimetallic Cooperation: A Mechanistic and Computational Study

By: Wu, Chen; et al

ACS Catalysis (2020), 10(14), 7934-7944.

1.1 **Catalysts:** Palladium diacetate, 2'-(Dicyclohexylphosphino)-*N,N*-dimethyl[1,1'-biphenyl]-2-amine

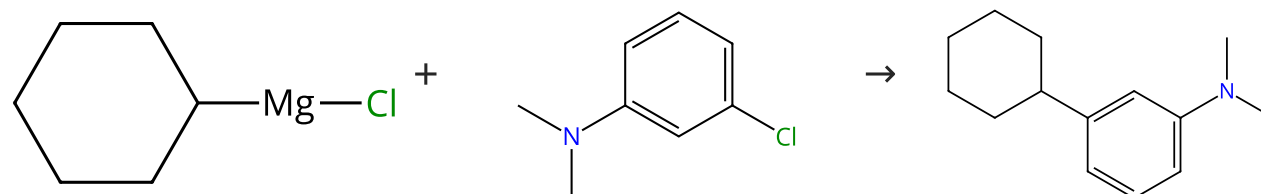
Solvents: 1,2-Dimethoxyethane; 30 min, rt

1.2 **Solvents:** Diethyl ether; rt; 4 h, 50 °C; 50 °C → rt

1.3 **Reagents:** Water; rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (33)

 Suppliers (55)

 Suppliers (6)

31-176-CAS-20540054

Steps: 1 Yield: 100%

Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow

By: Wei, Xiao-Jing; et al

Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.

1.1 **Catalysts:** 1*H*-Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1)

Solvents: Tetrahydrofuran; 10 min, rt

1.2 **Catalysts:** Iron(III) acetylacetonate

Solvents: Tetrahydrofuran; 5 min, 25 °C

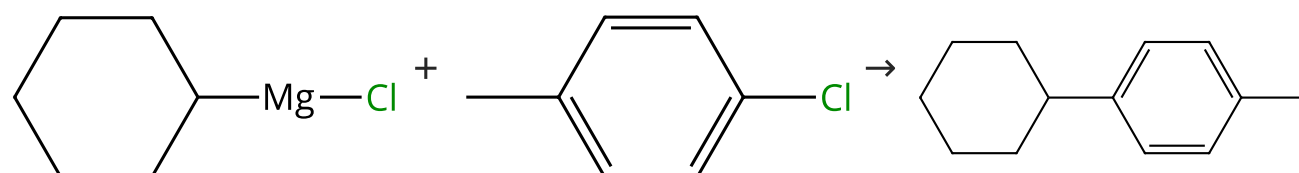
1.3 **Reagents:** Hydrochloric acid

Solvents: Water

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (33)

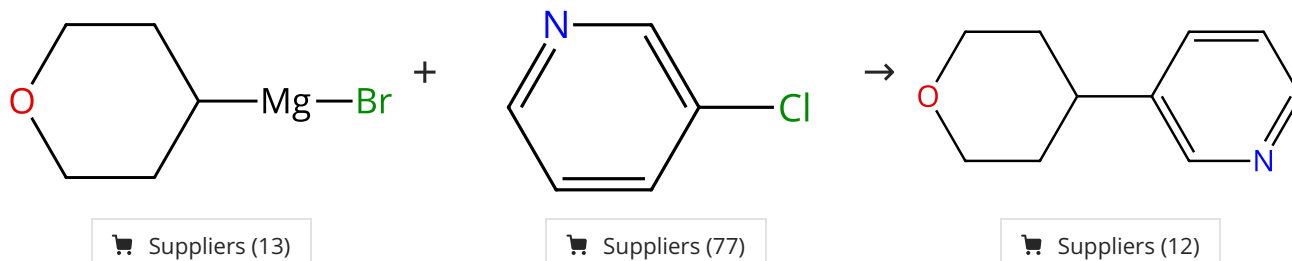
 Suppliers (82)

 Suppliers (46)

31-176-CAS-20540049	Steps: 1 Yield: 100%	Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow By: Wei, Xiao-Jing; et al Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.
1.1 Catalysts: 1 <i>H</i> -Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1) Solvents: Tetrahydrofuran; 10 min, rt		
1.2 Catalysts: Iron(III) acetylacetonate Solvents: Tetrahydrofuran; 5 min, 25 °C		
1.3 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 4 (1 Reaction)

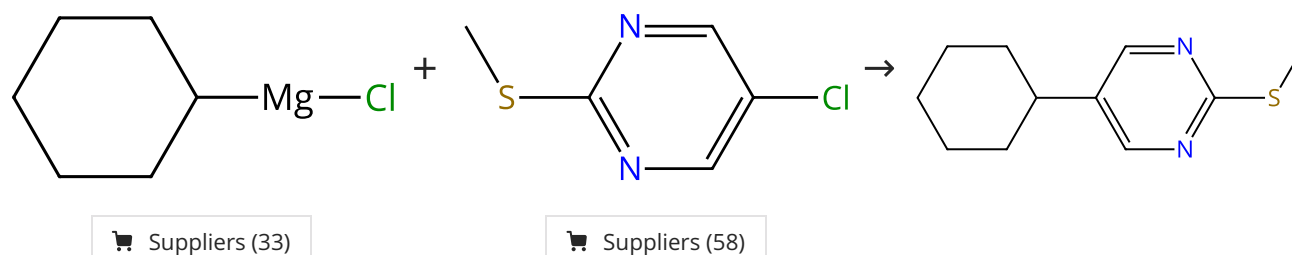
Steps: 1 Yield: 100%



31-176-CAS-20540068	Steps: 1 Yield: 100%	Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow By: Wei, Xiao-Jing; et al Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.
1.1 Reagents: Bromo(1-methylethyl)magnesium Catalysts: 1 <i>H</i> -Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1) Solvents: Tetrahydrofuran; 10 min, rt		
1.2 Catalysts: Iron(III) acetylacetonate Solvents: Tetrahydrofuran; 20 min, 25 °C		
1.3 Reagents: Ammonia Solvents: Water		
Experimental Protocols		

Scheme 5 (1 Reaction)

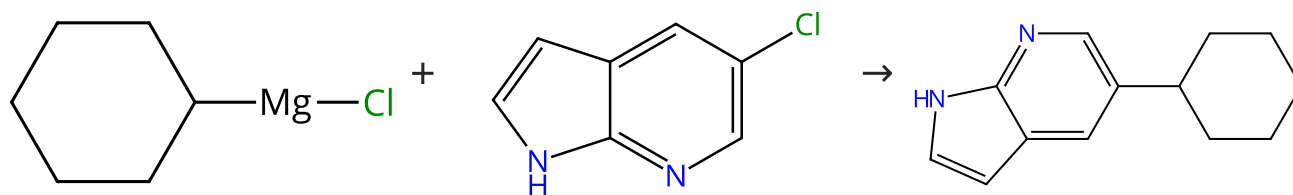
Steps: 1 Yield: 100%



31-176-CAS-20540071	Steps: 1 Yield: 100%	Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow By: Wei, Xiao-Jing; et al Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.
1.1 Catalysts: 1 <i>H</i> -Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1) Solvents: Tetrahydrofuran; 10 min, rt		
1.2 Catalysts: Iron(III) acetylacetonate Solvents: Tetrahydrofuran; 1 min, 25 °C		
1.3 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (33)

Suppliers (91)

Suppliers (4)

31-176-CAS-20540057

Steps: 1 Yield: 100%

Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow

By: Wei, Xiao-Jing; et al

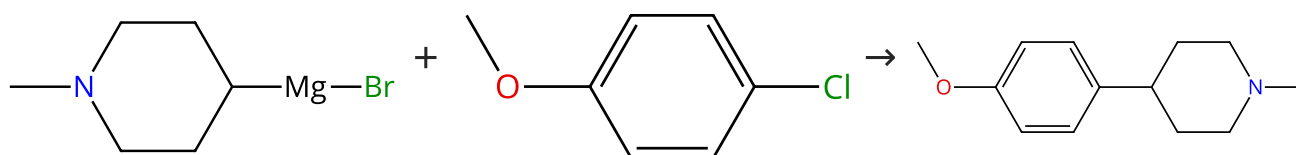
Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.

- 1.1 **Catalysts:** 1*H*-Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1)
Solvents: Tetrahydrofuran; 10 min, rt
- 1.2 **Catalysts:** Iron(III) acetylacetonate
Solvents: Tetrahydrofuran; 5 min, 25 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (72)

Suppliers (5)

31-176-CAS-20540069

Steps: 1 Yield: 100%

Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow

By: Wei, Xiao-Jing; et al

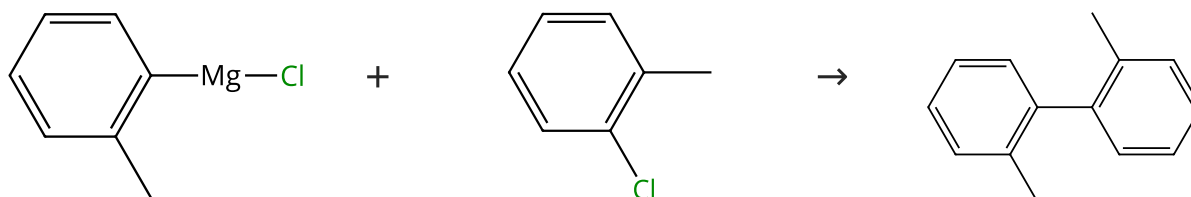
Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.

- 1.1 **Reagents:** Bromo(1-methylethyl)magnesium
Catalysts: 1*H*-Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1)
Solvents: Tetrahydrofuran; 10 min, rt
- 1.2 **Catalysts:** Iron(III) acetylacetonate
Solvents: Tetrahydrofuran; 20 min, 25 °C
- 1.3 **Reagents:** Ammonia
Solvents: Water

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (15)

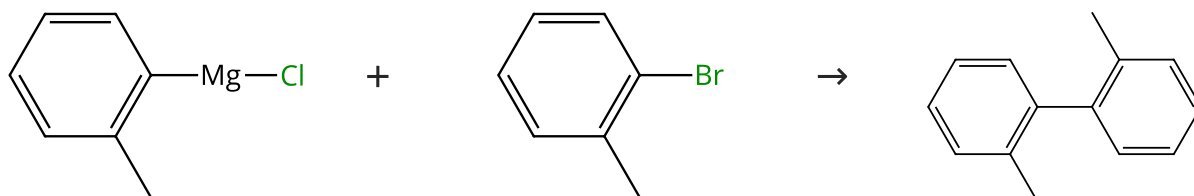
Suppliers (71)

Suppliers (57)

31-176-CAS-2695245 Steps: 1 Yield: 100% 1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 1<i>H</i>-Imidazolium, 1-(phenylmethyl)-3-(1,21,23,25-tetrapentyl-2,20:3,19-dimetheno-1<i>H</i>,21<i>H</i>,23<i>H</i>,25<i>H</i>-bis[1,3]dioxocino[5,4-<i>i</i>:5',4'-<i>i'</i>]benzo[1,2-<i>d</i>:5,4-<i>d'</i>]bis[1,3]benzodioxocin-7-yl)-, bromide (1:1) Solvents: Tetrahydrofuran, 1,4-Dioxane; 24 h, 50 °C	Subtle Steric Effects in Nickel-Catalysed Kumada-Tamao-Corriu Cross-Coupling Using Resorcinarenyl-Imidazolium Salts By: Sahin, Neslihan; et al European Journal of Organic Chemistry (2013), 2013(20), 4443-4449.
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Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (15)

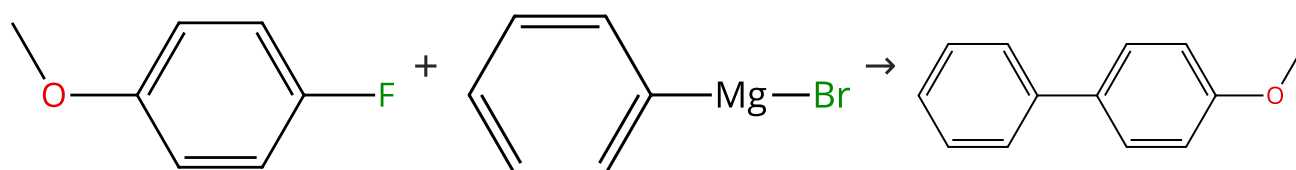
Suppliers (60)

Suppliers (57)

31-176-CAS-527039 Steps: 1 Yield: 100% 1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 1<i>H</i>-Imidazolium, 1-(phenylmethyl)-3-(1,21,23,25-tetrapentyl-2,20:3,19-dimetheno-1<i>H</i>,21<i>H</i>,23<i>H</i>,25<i>H</i>-bis[1,3]dioxocino[5,4-<i>i</i>:5',4'-<i>i'</i>]benzo[1,2-<i>d</i>:5,4-<i>d'</i>]bis[1,3]benzodioxocin-7-yl)-, bromide (1:1) Solvents: Tetrahydrofuran, 1,4-Dioxane; 2 h, 100 °C	Subtle Steric Effects in Nickel-Catalysed Kumada-Tamao-Corriu Cross-Coupling Using Resorcinarenyl-Imidazolium Salts By: Sahin, Neslihan; et al European Journal of Organic Chemistry (2013), 2013(20), 4443-4449.
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Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

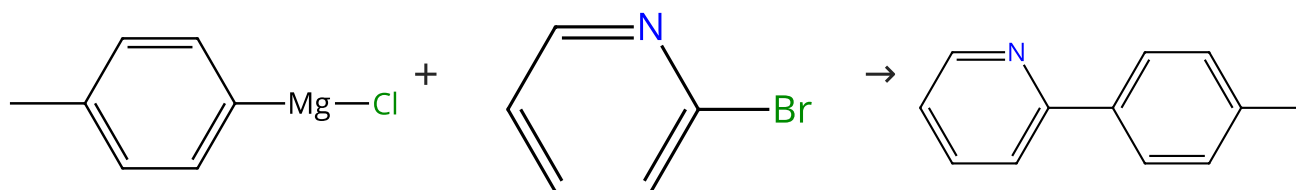
Suppliers (55)

Suppliers (69)

31-176-CAS-22542836 Steps: 1 Yield: 100% 1.1 Catalysts: Palladium diacetate, 2-(Dicyclohexylphosphino)biphenyl Solvents: 1,2-Dimethoxyethane; 30 min, rt 1.2 Solvents: Diethyl ether; rt; 4 h, 50 °C; 50 °C → rt 1.3 Reagents: Water; rt	Aryl Fluoride Activation through Palladium-Magnesium Bimetallic Cooperation: A Mechanistic and Computational Study By: Wu, Chen; et al ACS Catalysis (2020), 10(14), 7934-7944.
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Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (9)

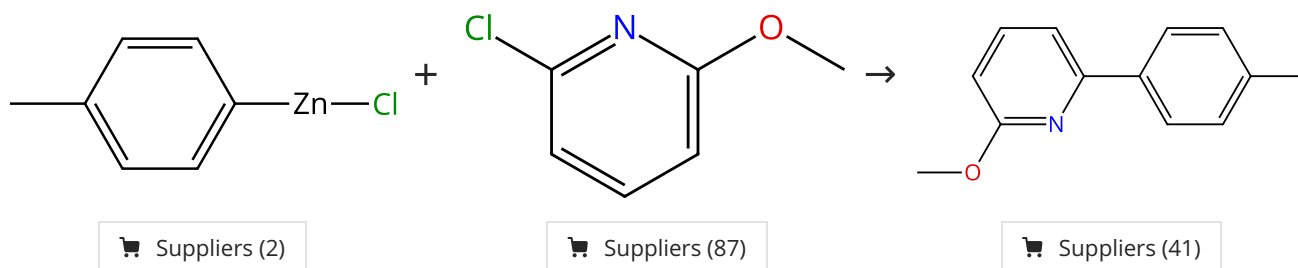
Suppliers (87)

Suppliers (70)

31-176-CAS-3369656	Steps: 1 Yield: 99%	Synthesis of Mixed Silylene-Carbene Chelate Ligands from N-Heterocyclic Silylcarbenes Mediated by Nickel
1.1 Catalysts: 1658479-98-1 Solvents: Tetrahydrofuran; 24 h, rt → 70 °C		By: Tan, Gengwen; et al
Experimental Protocols		Angewandte Chemie, International Edition (2015), 54(7), 2214-2218.

Scheme 12 (1 Reaction)

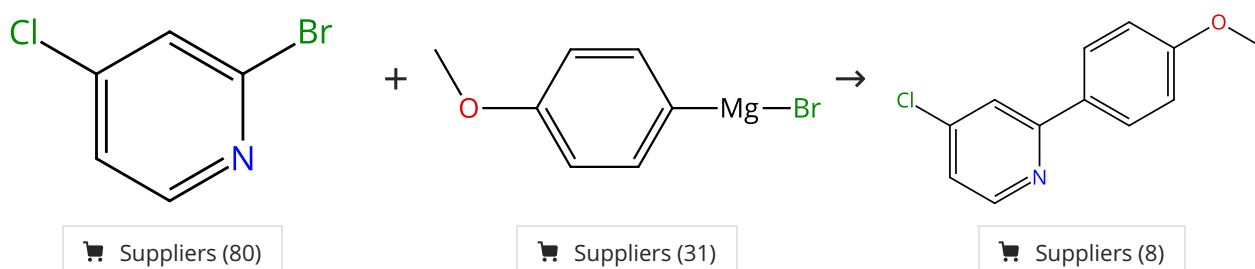
Steps: 1 Yield: 99%



31-176-CAS-883775	Steps: 1 Yield: 99%	P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> ² -[2-(diphenylphosphino-κ <i>P</i>)phenyl]- <i>N</i> ¹ , <i>N</i> ¹ -dimethyl-1,2-benzenediaminato-κ <i>N</i> ¹ ,κ <i>N</i> ²]nickel Solvents: Tetrahydrofuran, <i>N</i> -Methyl-2-pyrrolidone; 24 h, 80 °C		By: Wu, Dan; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.

Scheme 13 (1 Reaction)

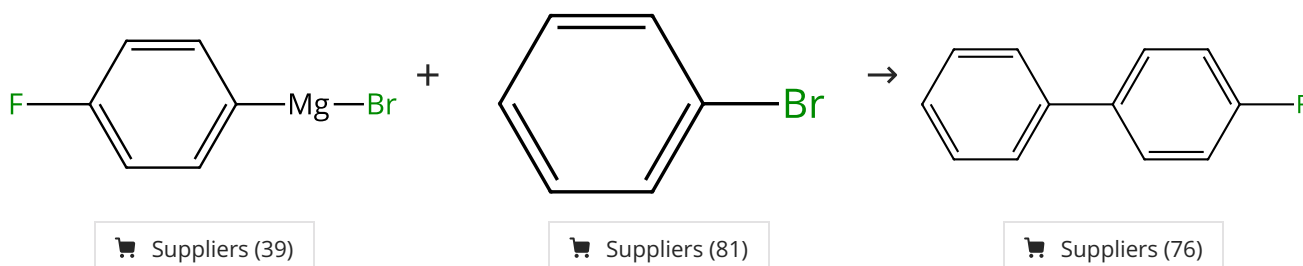
Steps: 1 Yield: 99%



31-614-CAS-32972820	Steps: 1 Yield: 99%	Unconventional Site Selectivity in Palladium-Catalyzed Cross-Couplings of Dichloroheteroarenes under Ligand-Controlled and Ligand-Free Systems
1.1 Catalysts: Palladium diacetate, 1,1-Bis(diphenylphosphino)ferrocene Solvents: Tetrahydrofuran; 2 h, 25 °C		By: Norman, Jacob P.; et al
		Journal of Organic Chemistry (2022), 87(11), 7414-7421.

Scheme 14 (1 Reaction)

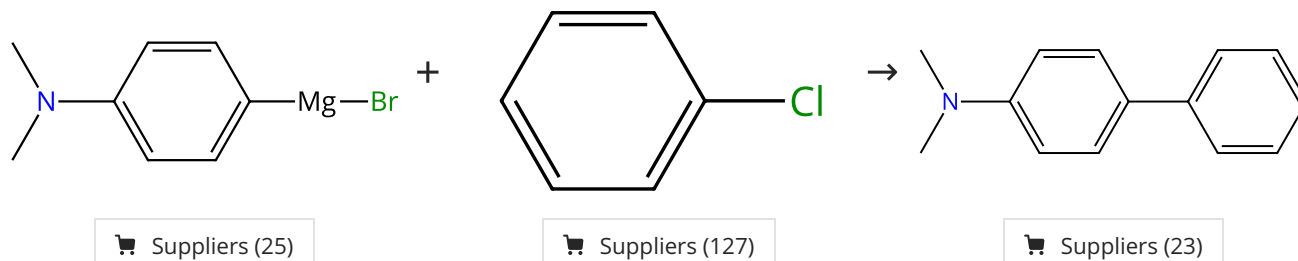
Steps: 1 Yield: 99%



31-176-CAS-18718059	Steps: 1 Yield: 99%	A Highly Active Cobalt Catalyst System for Kumada Biaryl Cross-Coupling By: Wu, Wenqin; et al Synlett (2018), 29(9), 1199-1202.
1.1 Catalysts: Sodium <i>tert</i> -butoxide, Cobalt chloride (CoCl ₂), 1 <i>H</i> -Imidazolium, 4,5-dihydro-1,3-bis(2,4,6-trimethylphenyl)-, chloride (1:1) Solvents: Tetrahydrofuran; 1 h, rt		
1.2 Solvents: Tetrahydrofuran; 1 h, rt		
1.3 Reagents: Ammonium chloride Solvents: Water; rt		
Experimental Protocols		

Scheme 15 (1 Reaction)

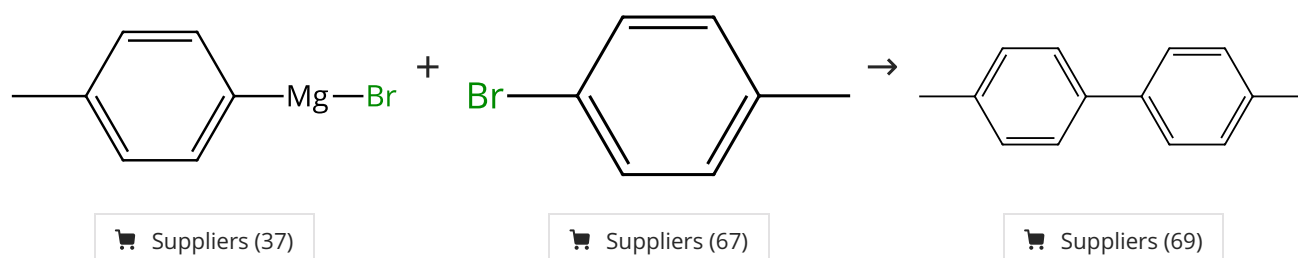
Steps: 1 Yield: 99%



31-176-CAS-13807714	Steps: 1 Yield: 99%	Amido pincer nickel catalyzed Kumada cross-coupling of aryl, heteroaryl, and vinyl chlorides By: Zhang, Xue-Qi; et al Synlett (2013), 24(16), 2081-2084.
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> -[2-(diphenylphosphino- κP)phenyl]- <i>N'</i> -(4-methylphenyl)benzenecarboximidamidato- $\kappa N, \kappa N'$]nickel Solvents: Tetrahydrofuran; 30 °C; 24 h, 30 °C		
1.2 Reagents: Acetic acid, Water		
Experimental Protocols		

Scheme 16 (1 Reaction)

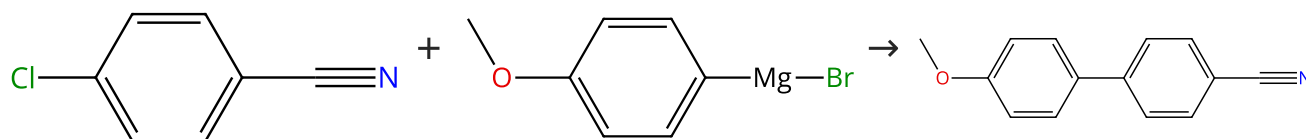
Steps: 1 Yield: 99%



31-176-CAS-9705416	Steps: 1 Yield: 99%	Formation of homoleptic tetracarbene versus cis-chelating dicarbene complexes of nickel(II) and applications in Kumada-Corriu couplings By: Vinh Huynh, Han; et al European Journal of Inorganic Chemistry (2009), (13), 1926-1931.
1.1 Catalysts: (<i>SP</i> -4-2)-Dibromo[1,3-propanediylbis(3-methyl-1 <i>H</i> -benzimidazol-1-yl-2(3 <i>H</i>)-ylidene)]nickel Solvents: Tetrahydrofuran; 12 h, rt		
Experimental Protocols		

Scheme 17 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (31)

Suppliers (48)

31-176-CAS-16236825

Steps: 1 Yield: 99%

Inherent vs Apparent Chemoselectivity in the Kumada-Corriu Cross-Coupling Reaction

By: Hua, XiYe; et al

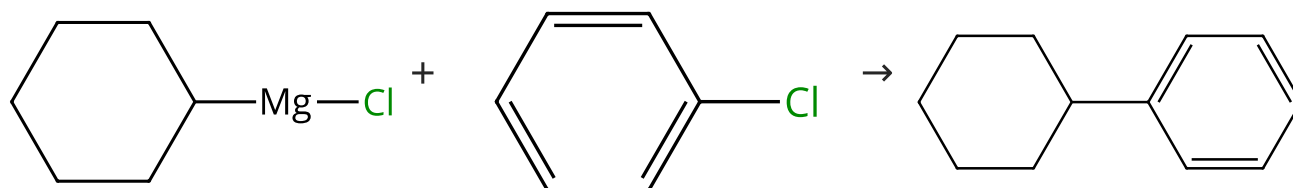
Organic Letters (2016), 18(20), 5312-5315.

- 1.1 **Catalysts:** Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Toluene; rt
- 1.2 **Solvents:** Tetrahydrofuran; rt
- 1.3 90 min, rt; 5 min, rt
- 1.4 **Reagents:** Ammonium chloride
Solvents: Water; rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (33)

Suppliers (127)

Suppliers (79)

31-176-CAS-20540044

Steps: 1 Yield: 99%

Visible-Light-Promoted Iron-Catalyzed C(sp²)-C(sp³) Kumada Cross-Coupling in Flow

By: Wei, Xiao-Jing; et al

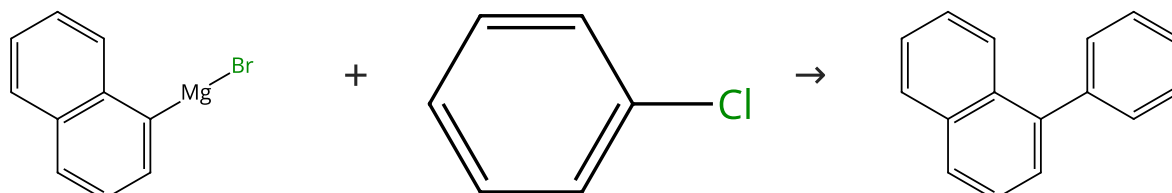
Angewandte Chemie, International Edition (2019), 58(37), 13030-13034.

- 1.1 **Catalysts:** 1*H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1)
Solvents: Tetrahydrofuran; 10 min, rt
- 1.2 **Catalysts:** Iron(III) acetylacetonate
Solvents: Tetrahydrofuran; 5 min, 25 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (25)

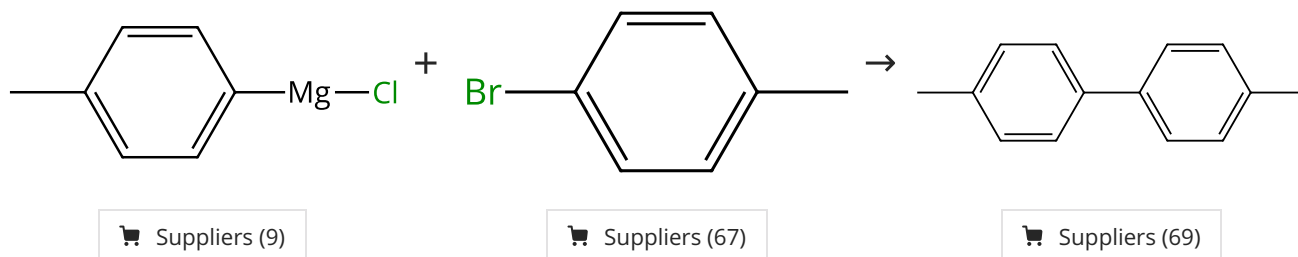
Suppliers (127)

Suppliers (68)

31-176-CAS-15991710	Steps: 1 Yield: 99%	Nickel(II) complexes of amine functionalized N-heterocyclic carbenes (NHCs), synthesis and catalysis in Kumada coupling of aryl chlorides
1.1 Catalysts: 1777781-30-2 Solvents: Tetrahydrofuran; 24 h, rt → 40 °C		By: Zhang, Shumiao; et al Journal of Organometallic Chemistry (2016), 820, 41-45.

Scheme 20 (1 Reaction)

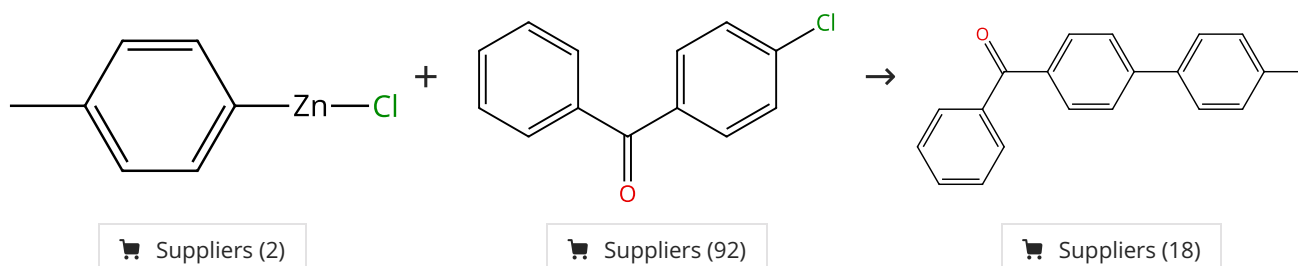
Steps: 1 Yield: 99%



31-176-CAS-14226052	Steps: 1 Yield: 99%	Synthesis of Mixed Silylene-Carbene Chelate Ligands from N-Heterocyclic Silylcarbenes Mediated by Nickel
1.1 Catalysts: 1658479-98-1 Solvents: Tetrahydrofuran; 24 h, rt → 70 °C		By: Tan, Gengwen; et al Angewandte Chemie, International Edition (2015), 54(7), 2214-2218.
Experimental Protocols		

Scheme 21 (1 Reaction)

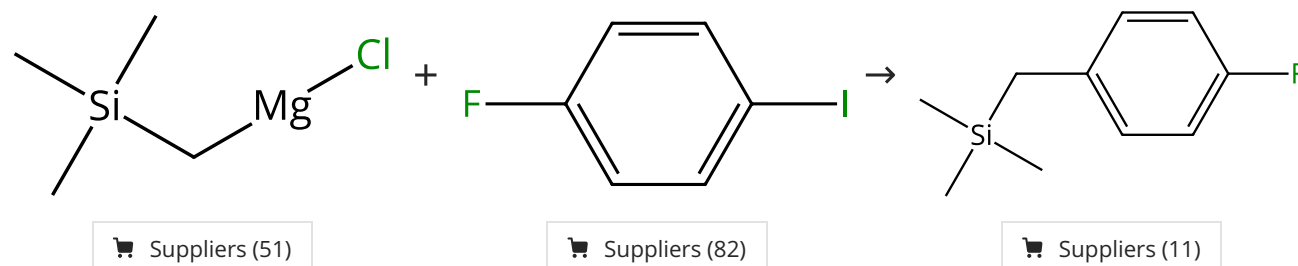
Steps: 1 Yield: 99%



31-176-CAS-6462746	Steps: 1 Yield: 99%	P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> ² -[2-(diphenylphosphino- κ <i>P</i>)phenyl]- <i>N</i> ¹ , <i>N</i> ¹ -dimethyl-1,2-benzenediaminato- κ <i>N</i> ¹ , κ <i>N</i> ²]nickel Solvents: Tetrahydrofuran, <i>N</i> -Methyl-2-pyrrolidone; 24 h, 80 °C		By: Wu, Dan; et al Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.
Experimental Protocols		

Scheme 22 (1 Reaction)

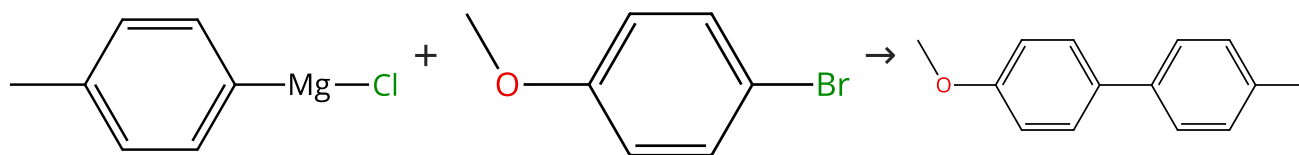
Steps: 1 Yield: 99%



31-176-CAS-14232118	Steps: 1 Yield: 99%	Catalysis of Kumada-Tamao-Corriu coupling by a (P ^{OCOP})Rh pincer complex
1.1 Catalysts: (SP-5-43)-[2,6-Bis[[bis(1-methylethyl)phosphino-κP]oxy]phenyl-κC]chlorohydorrhodium Solvents: Diethyl ether; 5 min, 22 °C		By: Timpa, Samuel D.; et al Dalton Transactions (2011), 40(20), 5426-5429.
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (9)

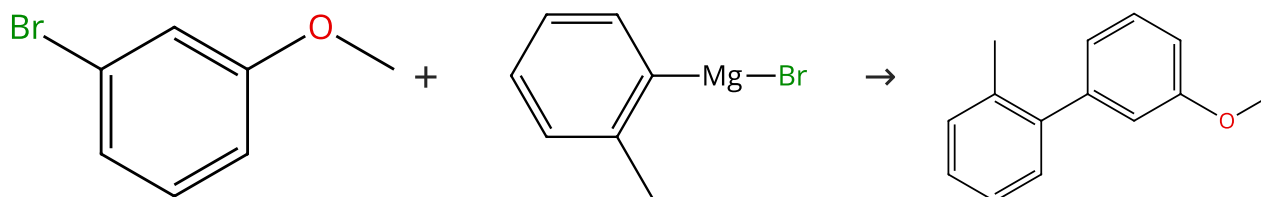
Suppliers (71)

Suppliers (43)

31-176-CAS-9969143	Steps: 1 Yield: 99%	Synthesis of Mixed Silylene-Carbene Chelate Ligands from N-Heterocyclic Silylcarbenes Mediated by Nickel
1.1 Catalysts: 1658479-98-1 Solvents: Tetrahydrofuran; 24 h, rt → 70 °C		By: Tan, Gengwen; et al Angewandte Chemie, International Edition (2015), 54(7), 2214-2218.
Experimental Protocols		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

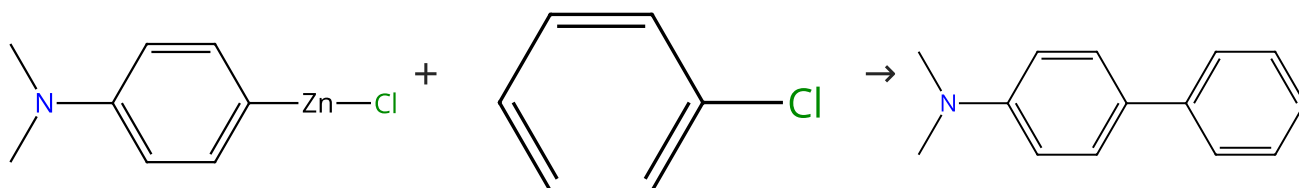
Suppliers (34)

Suppliers (11)

31-176-CAS-5352519	Steps: 1 Yield: 99%	Pyrazole-tethered phosphine ligands for Pd(0): useful catalysts for Stille, Kumada and Hiyama cross-coupling reactions
1.1 Catalysts: Tris(dibenzylideneacetone)dipalladium, 3-(1,1-Dimethylethyl)-1-[2-(diphenylphosphino)phenyl]-5-methyl-1H-pyrazole Solvents: Toluene, Tetrahydrofuran; rt; 10 h, 55 °C		By: Pal, Anwasha; et al Tetrahedron (2010), 66(29), 5451-5458.
1.2 Reagents: Water; rt		
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

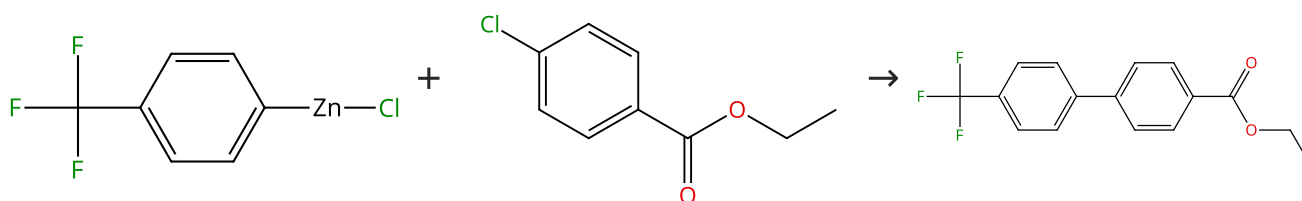
Suppliers (127)

Suppliers (23)

31-176-CAS-10503150	Steps: 1 Yield: 99%	P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> ² -[2-(diphenylphosphino-κ <i>P</i>)phenyl]- <i>N</i> ¹ , <i>N</i> ¹ -dimethyl-1,2-benzenediaminato-κ <i>N</i> ¹ ,κ <i>N</i> ²]nickel Solvents: Tetrahydrofuran, <i>N</i> -Methyl-2-pyrrolidone; 24 h, 80 °C	By: Wu, Dan; et al Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.	
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

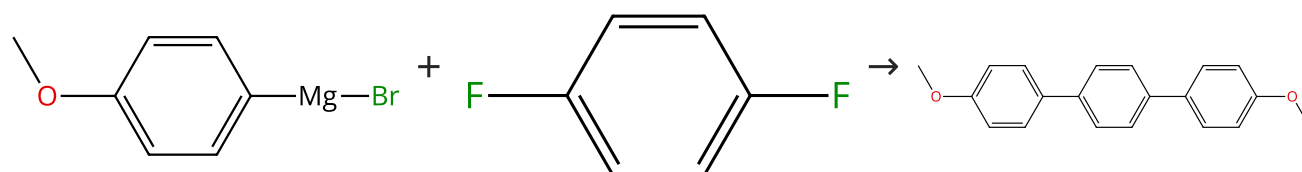
Suppliers (71)

Suppliers (13)

31-176-CAS-13613817	Steps: 1 Yield: 99%	P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> ² -[2-(diphenylphosphino-κ <i>P</i>)phenyl]- <i>N</i> ¹ , <i>N</i> ¹ -dimethyl-1,2-benzenediaminato-κ <i>N</i> ¹ ,κ <i>N</i> ²]nickel Solvents: Tetrahydrofuran, <i>N</i> -Methyl-2-pyrrolidone; 24 h, 80 °C	By: Wu, Dan; et al Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.	
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (31)

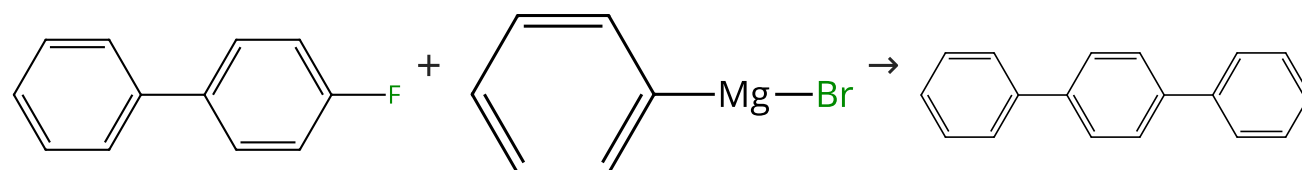
Suppliers (67)

Suppliers (38)

31-176-CAS-10741122	Steps: 1 Yield: 99%	Cross-Coupling of ArX with ArMgBr Catalyzed by N-Heterocyclic Carbene-Based Nickel Complexes
1.1 Catalysts: (<i>SP</i> -4-3)-Bromo[<i>N</i> -[2-[2,3-dihydro-3-(phenylmethyl)-1 <i>H</i> -imidazol-1-yl-κ <i>C</i> ²]phenyl]- <i>N</i> -(4-methylphenyl)- <i>P,P</i> -diphenylphosphinimidic amidato(3-)-κ <i>N</i> ,κ <i>N</i>]nickel Solvents: Tetrahydrofuran; 25 °C; 12 h, 25 °C	By: Guo, Wang-Jun; et al Journal of Organic Chemistry (2013), 78(3), 1054-1061.	
1.2 Reagents: Water		
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (76)

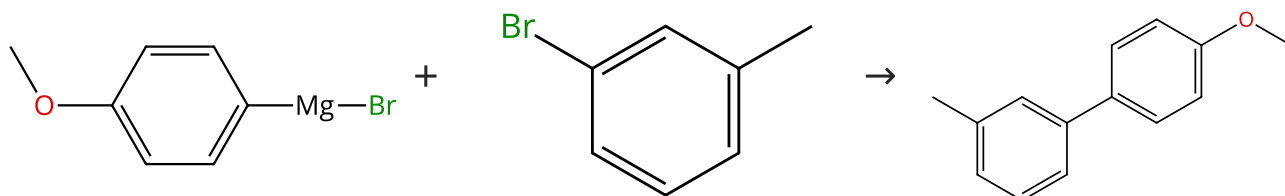
Suppliers (55)

Suppliers (113)

31-176-CAS-22542838	Steps: 1 Yield: 99%	Aryl Fluoride Activation through Palladium-Magnesium Bimetallic Cooperation: A Mechanistic and Computational Study By: Wu, Chen; et al ACS Catalysis (2020), 10(14), 7934-7944.
1.1	Catalysts: Palladium diacetate, 2'-(Dicyclohexylphosphino)- <i>N,N</i> -dimethyl[1,1'-biphenyl]-2-amine Solvents: Tetrahydrofuran; rt; 16 h, 80 °C	

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (31)

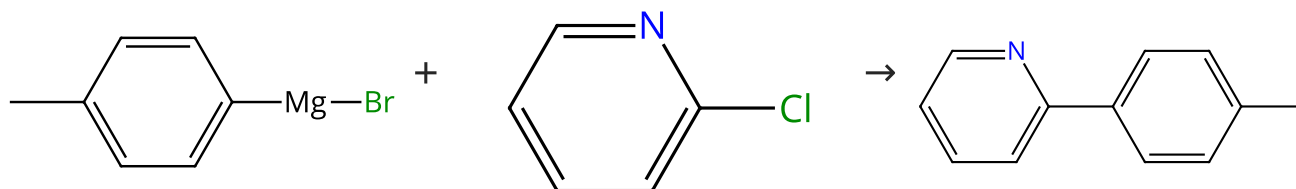
Suppliers (58)

Suppliers (37)

31-176-CAS-14205446	Steps: 1 Yield: 99%	Pyrazole-tethered phosphine ligands for Pd(0): useful catalysts for Stille, Kumada and Hiyama cross-coupling reactions By: Pal, Anwasha; et al Tetrahedron (2010), 66(29), 5451-5458.
1.1	Catalysts: Tris(dibenzylideneacetone)dipalladium, 3-(1,1-Dimethylethyl)-1-[2-(diphenylphosphino)phenyl]-5-methyl-1 <i>H</i> -pyrazole Solvents: Toluene, Tetrahydrofuran; rt; 10 h, 55 °C	
1.2	Reagents: Water; rt	
Experimental Protocols		

Scheme 30 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (37)

Suppliers (66)

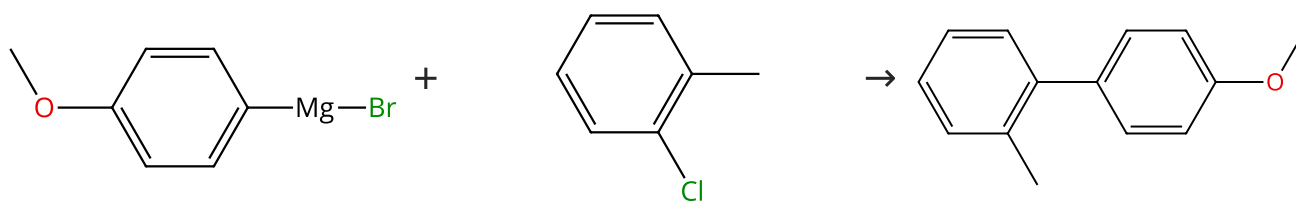
Suppliers (70)

31-176-CAS-11671206	Steps: 1 Yield: 99%	Trinuclear copper(I) complex of 1,3-bis(2-pyridinylmethyl)imidazolyldiene as a carbene-transfer reagent for the preparation of catalytically active nickel(II) and palladium(II) complexes By: Chen, Chao; et al Journal of Organometallic Chemistry (2012), 696(26), 4166-4172.
1.1	Catalysts: Nickel(1+), chloro[1,3-dihydro-1,3-bis[(2-pyridinyl-κM)methyl]-2 <i>H</i> -imidazol-2-ylidene-κC]-, (<i>SP</i> -4-3)-, hexafluorophosphate(1-)(1:1) Solvents: Tetrahydrofuran; 12 h, rt	

31-176-CAS-15064831	Steps: 1 Yield: 99%	Kumada Coupling of Aryl, Heteroaryl, and Vinyl Chlorides Catalyzed by Amido Pincer Nickel Complexes By: Liu, Ning; et al Journal of Organic Chemistry (2011), 76(24), 10031-10038.
1.1	Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> -[2-(diphenylphosphino-κ <i>P</i>)phenyl]-α-(diphenylphosphinyl-κ <i>O</i>)benzenemethanaminato-κM]nickel Solvents: Diethyl ether; 25 °C; 12 h, 25 °C	
1.2	Solvents: Water; 25 °C	
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (31)

Suppliers (71)

Suppliers (41)

31-176-CAS-4169868

Steps: 1 Yield: 99%

Cross-Coupling of ArX with ArMgBr Catalyzed by N-Heterocyclic Carbene-Based Nickel Complexes

By: Guo, Wang-Jun; et al

Journal of Organic Chemistry (2013), 78(3), 1054-1061.

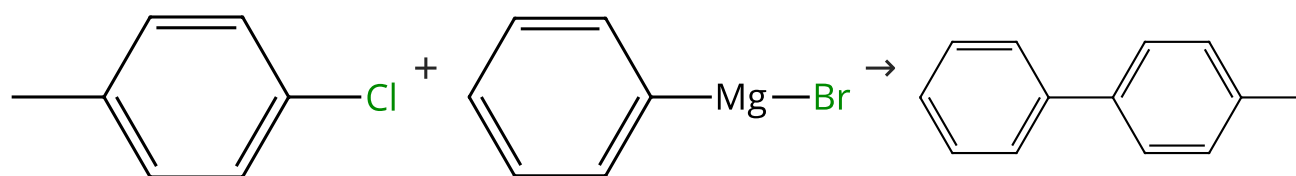
1.1 **Catalysts:** (*SP*-4-3)-Bromo[*N*-[2-[2,3-dihydro-3-(phenylmethyl)-1*H*-imidazol-1-yl-κC²]phenyl]-*N*-(4-methylphenyl)-*P,P*-diphenylphosphinimidic amidato(3-)-κ*N*,κ*N*]nickel
Solvents: Tetrahydrofuran; 25 °C; 17 h, 70 °C

1.2 **Reagents:** Water

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (55)

Suppliers (82)

31-176-CAS-1161253

Steps: 1 Yield: 99%

Synthesis, structure and reactivity of NHC-nickel complexes containing carbonato ligands

By: Guo, Jin; et al

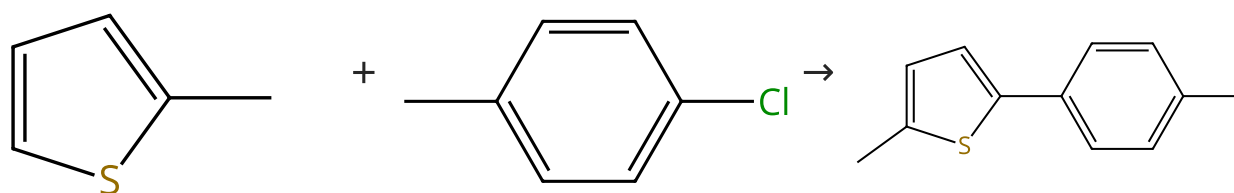
Inorganic Chemistry Communications (2013), 31, 74-78.

1.1 **Catalysts:** (*SP*-4-2)-[Carbonato(2-)-κ*O*,κ*O*][1,3-propanediylbis[3-(2,4,6-trimethylphenyl)-1*H*-imidazol-1-yl-2(3*H*)-ylidene]] nickel
Solvents: Tetrahydrofuran; 40 h, rt

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

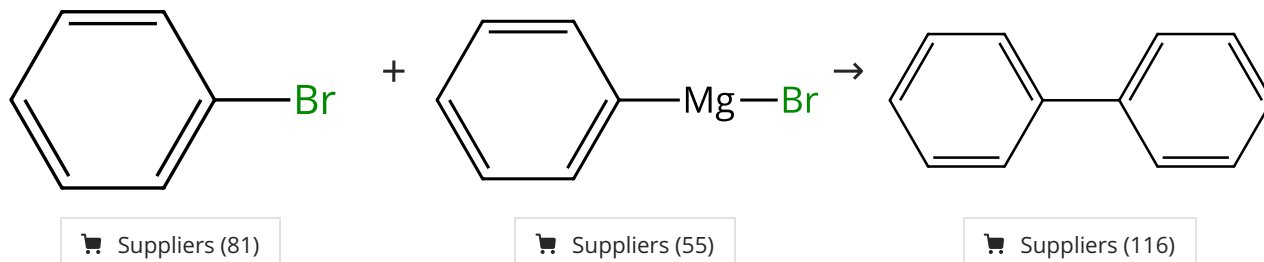
Suppliers (82)

Suppliers (10)

31-176-CAS-8342136	Steps: 1 Yield: 99%	Generation of metalated thiophenes with Grignard reagent and catalytic secondary amine for the cross coupling reaction with aryl halides
1.1 Reagents: Ethylmagnesium chloride Catalysts: Dicyclohexylamine Solvents: Tetrahydrofuran; 24 h, 60 °C		By: Tanaka, Shota; et al
1.2 Catalysts: (<i>SP</i> -4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro(3-chloropyridine- κ M)palladium; 24 h, 60 °C		Tetrahedron Letters (2012), 53(9), 1173-1176.
1.3 Reagents: Ammonium chloride Solvents: Water; rt		
Experimental Protocols		

Scheme 34 (1 Reaction)

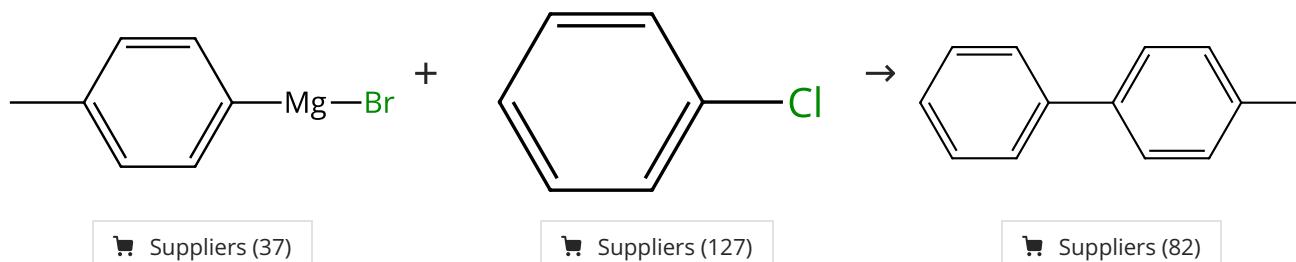
Steps: 1 Yield: 99%



31-176-CAS-12161617	Steps: 1 Yield: 99%	Formation of homoleptic tetracarbene versus cis-chelating dicarbene complexes of nickel(II) and applications in Kumada-Corriu couplings
1.1 Catalysts: (<i>SP</i> -4-2)-Dibromo[1,3-propanediylbis(3-methyl-1 <i>H</i> -benzimidazol-1-yl-2(3 <i>H</i>)-ylidene)]nickel Solvents: Tetrahydrofuran; 12 h, rt		By: Vinh Huynh, Han; et al
Experimental Protocols		European Journal of Inorganic Chemistry (2009), (13), 1926-1931.

Scheme 35 (2 Reactions)

Steps: 1 Yield: 99%

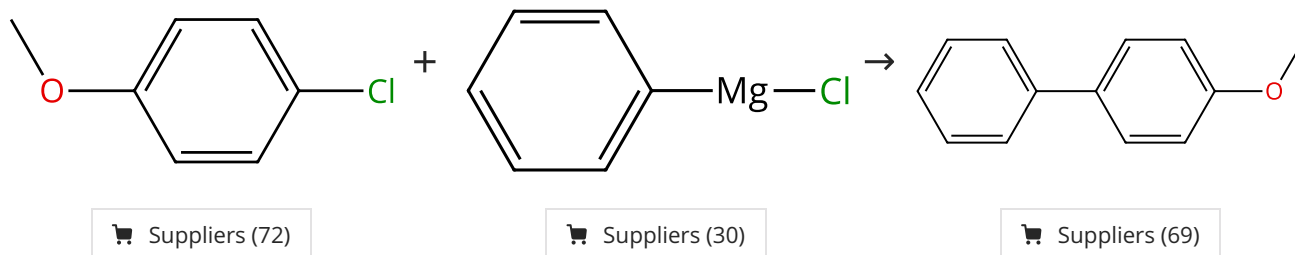


31-176-CAS-14896620	Steps: 1 Yield: 99%	Kumada Coupling of Aryl, Hetero aryl, and Vinyl Chlorides Catalyzed by Amido Pincer Nickel Complexes
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[α -(diphenylphosphino- κ P)- <i>N</i> -[2-(diphenylphosphinyl- κ O)phenyl]benzenemethanaminato- κ M]nickel Solvents: Tetrahydrofuran; 25 °C; 24 h, 25 °C		By: Liu, Ning; et al
1.2 Solvents: Water; 25 °C		Journal of Organic Chemistry (2011), 76(24), 10031-10038.
Experimental Protocols		

31-176-CAS-13311937 Steps: 1 Yield: 99% 1.1 Catalysts: (<i>SP</i>-4-1)-<i>rel</i>[(<i>αR</i>)-<i>α</i>-[Bis(1-methylethyl)phosphino-<i>κP</i>]-<i>N</i>-[2-(3,5-diphenyl-1<i>H</i>-pyrazol-1-yl)phenyl]benzenemethanamine][(<i>αS</i>)-<i>α</i>-[bis(1-methylethyl)phosphino-<i>κP</i>]-<i>N</i>-[2-(3,5-diphenyl-1<i>H</i>-pyrazol-1-yl)phenyl]benzenemethanamine] dichloronickel Solvents: Diethyl ether; 25 °C; 12 h, 25 °C 1.2 Reagents: Water; 25 °C Experimental Protocols	Nickel-Catalyzed Cross-Coupling of Non-Activated or Functionalized Aryl Halides with Aryl Grignard Reagents By: Xie, Lan-Gui; et al Chemistry - A European Journal (2010), 16(34), 10332-10336, S10332/1-S10332/49.
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Scheme 36 (1 Reaction)

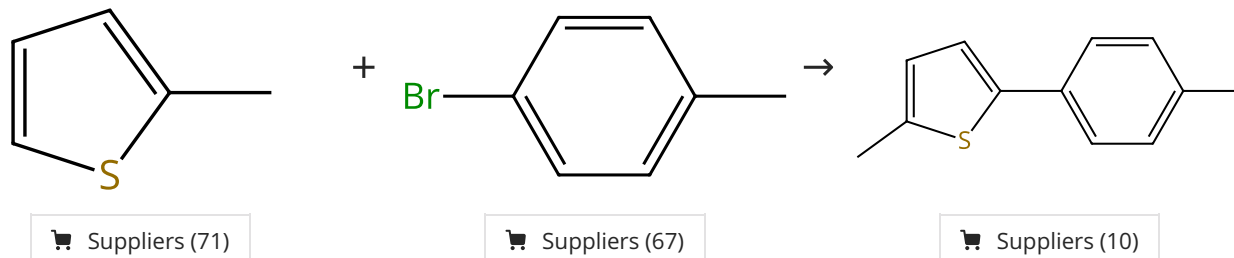
Steps: 1 Yield: 99%



31-176-CAS-12945506 Steps: 1 Yield: 99% 1.1 Catalysts: (<i>SP</i>-4-1)-Bis[2-[[2,3-dihydro-3-(phenylmethyl)-1<i>H</i>-imidazol-1-yl-<i>κC</i>²]-methyl]-1<i>H</i>-benzimidazolato(3-)-<i>κN</i>¹]nickel Solvents: Tetrahydrofuran; 20 min, rt	N-donor functionalized N-heterocyclic carbene nickel(II) complexes in the Kumada coupling By: Berding, Joris; et al Dalton Transactions (2009), (35), 6948-6955.
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Scheme 37 (1 Reaction)

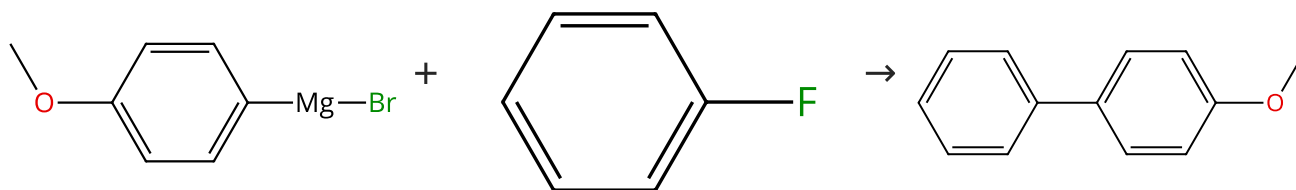
Steps: 1 Yield: 99%



31-176-CAS-651442 Steps: 1 Yield: 99% 1.1 Reagents: Ethylmagnesium chloride Catalysts: Dicyclohexylamine Solvents: Tetrahydrofuran; 24 h, 60 °C 1.2 Catalysts: (Bis(diphenylphosphino)ethane)dichloronickel; 24 h, rt 1.3 Reagents: Ammonium chloride Solvents: Water; rt Experimental Protocols	Generation of metalated thiophenes with Grignard reagent and catalytic secondary amine for the cross coupling reaction with aryl halides By: Tanaka, Shota; et al Tetrahedron Letters (2012), 53(9), 1173-1176.
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Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (31)

Suppliers (88)

Suppliers (69)

31-176-CAS-10562149

Steps: 1 Yield: 99%

Cross-Coupling of ArX with ArMgBr Catalyzed by N-Heterocyclic Carbene-Based Nickel Complexes

By: Guo, Wang-Jun; et al

Journal of Organic Chemistry (2013), 78(3), 1054-1061.

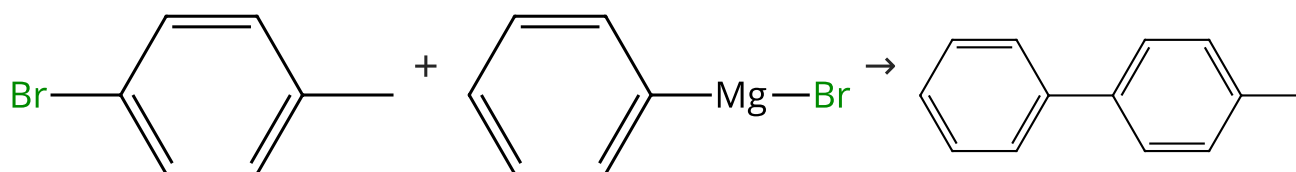
1.1 **Catalysts:** (SP-4-3)-Bromo[*N*-[2-[2,3-dihydro-3-(phenylmethyl)-1*H*-imidazol-1-yl-κC²]phenyl]-*N*-(4-methylphenyl)-*P,P*-diphenylphosphinimidic amidato(3-)-κ*N*,κ*N*]nickel
Solvents: Tetrahydrofuran; 25 °C; 17 h, 25 °C

1.2 **Reagents:** Water

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (67)

Suppliers (55)

Suppliers (82)

31-176-CAS-3549125

Steps: 1 Yield: 99%

Synthesis, structure and reactivity of NHC-nickel complexes containing carbonato ligands

By: Guo, Jin; et al

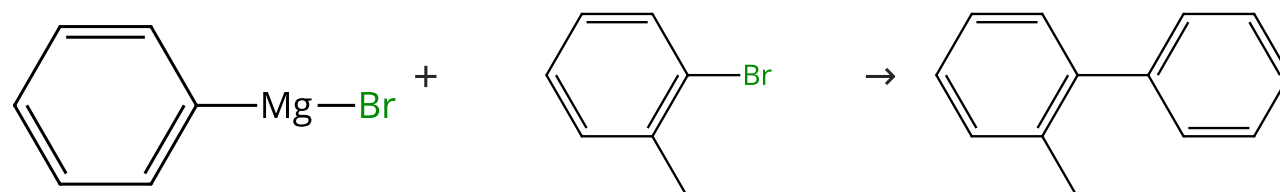
Inorganic Chemistry Communications (2013), 31, 74-78.

1.1 **Catalysts:** (SP-4-2)-[Carbonato(2-)-κ*O*,κ*O'*][1,3-propanediylbis[3-(2,4,6-trimethylphenyl)-1*H*-imidazol-1-yl-2(3*H*)-ylidene]]nickel
Solvents: Tetrahydrofuran; 40 h, rt

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (55)

Suppliers (60)

Suppliers (63)

31-176-CAS-5693742

Steps: 1 Yield: 99%

Synthesis, structure and reactivity of NHC-nickel complexes containing carbonato ligands

By: Guo, Jin; et al

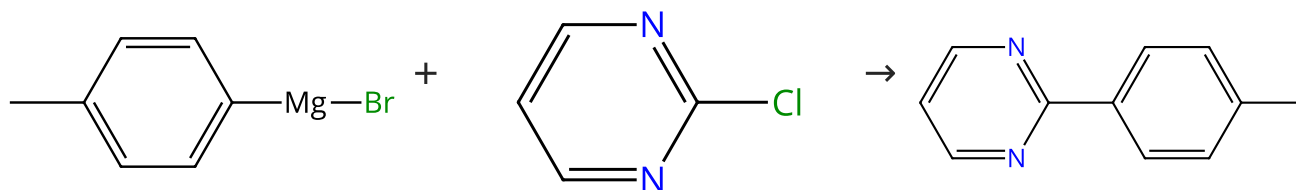
Inorganic Chemistry Communications (2013), 31, 74-78.

1.1 **Catalysts:** (SP-4-2)-[Carbonato(2-)-κ*O*,κ*O'*][1,3-propanediylbis[3-(2,4,6-trimethylphenyl)-1*H*-imidazol-1-yl-2(3*H*)-ylidene]]nickel
Solvents: Tetrahydrofuran; 40 h, rt

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (37)

Suppliers (87)

Suppliers (51)

31-176-CAS-13802652

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Nickel(1+), chloro[1,3-dihydro-1,3-bis[(2-pyridinyl-κM)methyl]-2H-imidazol-2-ylidene-κC]-, (SP-4-3)-, hexafluorophosphate(1-) (1:1)
Solvents: Tetrahydrofuran; 12 h, rt

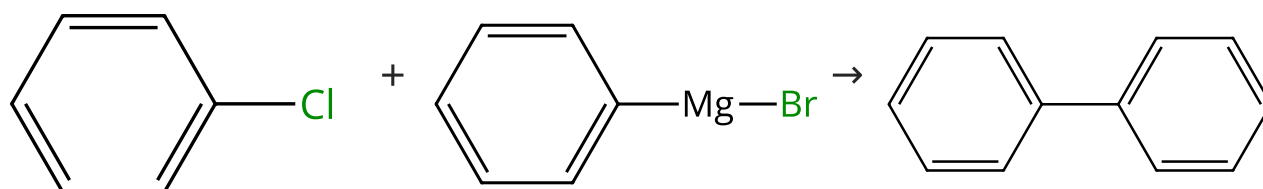
Trinuclear copper(I) complex of 1,3-bis(2-pyridinylmethyl)imidazolydene as a carbene-transfer reagent for the preparation of catalytically active nickel(II) and palladium(II) complexes

By: Chen, Chao; et al

Journal of Organometallic Chemistry (2012), 696(26), 4166-4172.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (127)

Suppliers (55)

Suppliers (116)

31-176-CAS-1421040

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** (SP-4-2)-[Carbonato(2-)-κO,κO'] [1,3-propanediylbis[3-(2,4,6-trimethylphenyl)-1H-imidazol-1-yl-2(3H)-ylidene]] nickel
Solvents: Tetrahydrofuran; 40 h, rt

Synthesis, structure and reactivity of NHC-nickel complexes containing carbonato ligands

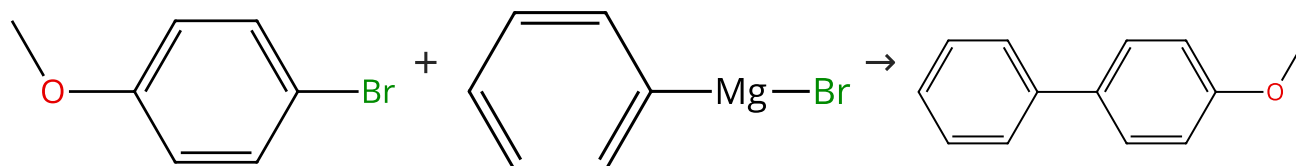
By: Guo, Jin; et al

Inorganic Chemistry Communications (2013), 31, 74-78.

Experimental Protocols

Scheme 43 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (71)

Suppliers (55)

Suppliers (69)

31-176-CAS-12694838

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** (SP-4-2)-[Carbonato(2-)-κO,κO'] [1,3-propanediylbis[3-(2,4,6-trimethylphenyl)-1H-imidazol-1-yl-2(3H)-ylidene]] nickel
Solvents: Tetrahydrofuran; 40 h, rt

Synthesis, structure and reactivity of NHC-nickel complexes containing carbonato ligands

By: Guo, Jin; et al

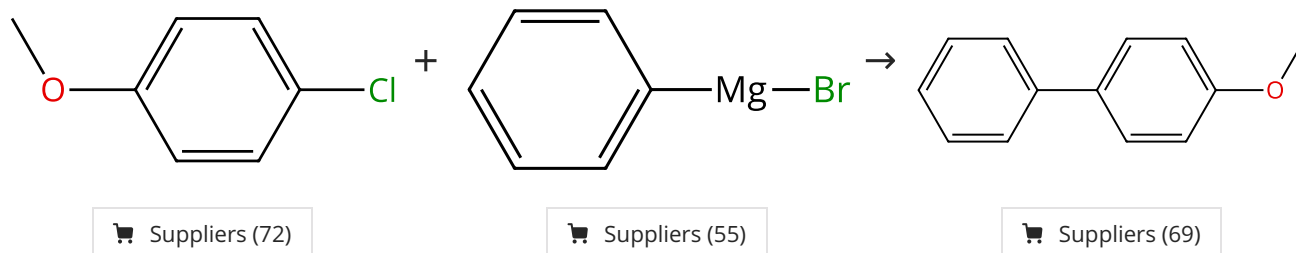
Inorganic Chemistry Communications (2013), 31, 74-78.

Experimental Protocols

31-176-CAS-1420524	Steps: 1 Yield: 99%	Pyrazole-tethered phosphine ligands for Pd(0): useful catalysts for Stille, Kumada and Hiyama cross-coupling reactions By: Pal, Anwasha; et al Tetrahedron (2010), 66(29), 5451-5458.
1.1 Catalysts: Tris(dibenzylideneacetone)dipalladium, 3-(1,1-Dimethylethyl)-1-[2-(diphenylphosphino)phenyl]-5-methyl-1 <i>H</i> -pyrazole Solvents: Toluene, Tetrahydrofuran; rt; 10 h, 55 °C		
1.2 Reagents: Water; rt		
Experimental Protocols		

Scheme 44 (3 Reactions)

Steps: 1 Yield: 99%



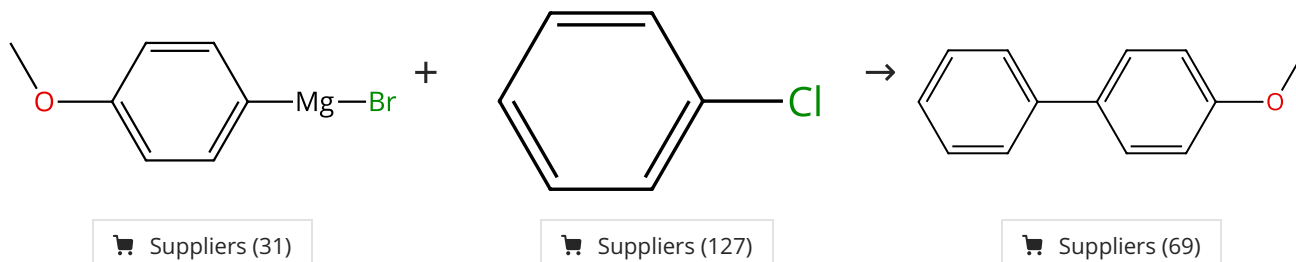
31-614-CAS-37241845	Steps: 1 Yield: 99%	Heterobidentate imidazo[1,5-a]pyridine and imidazo[1,5-a]quinoline N-heterocyclic carbene (NHC) ligands, catalyst complexes thereof, and methods using same Assignee: Rutgers, The State University of New Jersey World Intellectual Property Organization, WO2023122078 A1 2023-06-29 PatentPak available
1.1 Catalysts: 2946617-75-8 Solvents: Tetrahydrofuran, 1,4-Dioxane; rt; 16 h, 80 °C		

31-176-CAS-15991703	Steps: 1 Yield: 99%	Nickel(II) complexes of amine functionalized N-heterocyclic carbenes (NHCs), synthesis and catalysis in Kumada coupling of aryl chlorides By: Zhang, Shumiao; et al Journal of Organometallic Chemistry (2016), 820, 41-45.
1.1 Catalysts: 1989683-92-2 Solvents: Tetrahydrofuran; 24 h, rt → 40 °C		

31-176-CAS-12409165	Steps: 1 Yield: 99%	Synthesis, structure and reactivity of NHC-nickel complexes containing carbonate ligands By: Guo, Jin; et al Inorganic Chemistry Communications (2013), 31, 74-78.
1.1 Catalysts: (<i>SP</i> -4-2)-[Carbonato(2-)-κO,κO'] [1,3-propanediylbis [3-(2,4,6-trimethylphenyl)-1 <i>H</i> -imidazol-1-yl-2(3 <i>H</i>)-ylidene]] nickel Solvents: Tetrahydrofuran; 40 h, rt		
Experimental Protocols		

Scheme 45 (1 Reaction)

Steps: 1 Yield: 99%



31-176-CAS-14075782	Steps: 1 Yield: 99%	Amido pincer nickel catalyzed Kumada cross-coupling of aryl, heteroaryl, and vinyl chlorides By: Zhang, Xue-Qi; et al Synlett (2013), 24(16), 2081-2084.
1.1 Catalysts: (<i>SP</i> -4-4)-Chloro[<i>N</i> -[2-(diphenylphosphino-κ <i>P</i>)phenyl]- <i>N'</i> -(4-methylphenyl)benzenecarboximidamido-κ <i>N</i> , κ <i>N'</i>]nickel Solvents: Tetrahydrofuran; 30 °C; 24 h, 30 °C		
1.2 Reagents: Acetic acid, Water		
Experimental Protocols		